

Section 98 — Plug DP And Treewidth Frontiers

A frontier may need connectivity, not just occupancy

This lesson derives A–C. Problem D is editorial-only.

A. Transfer matrices for repeated profiles

The one-row domino transition is a matrix indexed by incoming and outgoing masks. Repeating an identical row H times is matrix exponentiation; the empty-to-empty entry is the answer. This separates local geometry from repetition and changes linear dependence on huge H into $O(\log H)$ matrix products.

B. Canonical connectivity labels

For connected cell subsets, each nonzero frontier label names a currently open connected component. Selecting a cell creates a label or merges its left and upper labels. Skipping a cell may make a label disappear; if another component remains open, the vanished component can never reconnect and the state is invalid. If it is the last component, mark the solution closed and prohibit future selections.

Relabel components by first appearance after every transition: $(7,0,7,4)$ becomes $(1,0,1,2)$. Labels are names, not information; canonicalization merges equivalent states.

C. Steiner subset merging

Let $dp[mask][v]$ be the minimum cost of a connected subgraph containing terminals in $mask$ and ending at v . Combine two proper submasks at the same v , then run multi-source Dijkstra to move the meeting point through graph edges. Every Steiner tree can be split at a branching vertex, while every transition joins connected valid subgraphs.

Boundary-design checklist

Ask which facts can still interact with unseen input. Occupancy needs bits; connectivity needs an equivalence relation; a bounded-width graph decomposition needs the choices on the active bag. Prove that forgotten objects have no edge or path to the future before dropping them.

When a component may disappear

Suppose the last frontier occurrence of label c is removed. No processed cell of that component touches the unprocessed region anymore. Therefore it can never connect to another component later. If the final object must be connected, the transition is legal only if that component already is the whole completed object and no future selected cell will be added. This is the central plug-DP forgetting proof.

Canonicalization should run after every merge or deletion. Scan the frontier left to right, map the first positive label to 1, the next unseen label to 2, and so on. Two histories that induce the same occupancy and equivalence relation must then obtain the same key.

Tree-decomposition transitions

At an introduce node, extend the bag state with every legal status of the new vertex. At a forget node, minimize or sum over its statuses only after checking all constraints that can no longer be inspected. At a join node, the two child subgraphs overlap exactly on the bag; combine compatible boundary states and correct any bag contribution counted twice.

Validate supplied decompositions: every graph vertex appears, every edge has a bag containing both endpoints, and bags containing one vertex form a connected subtree. For small graphs, compare with exhaustive subsets. For plug DP, compare canonical states themselves under mirrored boards to catch label-name leaks.

Exercises

A–C cover repeated, connected, and terminal-subset frontiers. D applies introduce/forget/join transitions on a supplied nice tree decomposition.